

PROJECT-7

Financial Decision Making for Stock Signals

Natural Language Processing

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Problem Statement



Task Definition

Given daily news articles, financial reports (10-K/10-Q), momentum signals, and price data for a stock, **predict whether to BUY, SELL, or HOLD** the stock for the next trading day.



Assets Covered



Bitcoin (BTC)
Cryptocurrency



Tesla (TSLA)
Equity Stock



Classification Problem

3-class classification under financial NLP domain:



BUY



HOLD



SELL



Key Challenges

- ✗ **Class imbalance** — HOLD dominates; BUY/SELL are rare
- ✗ **Noisy news text** — irrelevant articles dilute signal
- ✗ **Scale difference** — BTC (\$62K–\$124K) vs TSLA (\$302–\$489)
- ✗ **Data leakage risk** — future information must not contaminate training

Dataset Details

Source & Assets

Source	CLEF 2026 FinMMEval
Assets	BTC + TSLA
Date Range	Aug 2025 – Mar 2026

Statistics

Total Records	476 (238 × 2)
BTC Price Range	\$62,754 – \$124,798
TSLA Price Range	\$302.63 – \$489.88

Data Split

Training Set80%

190 records per asset

Validation Set10%

24 records per asset

Test Set10%

24 records per asset

i Chronological split — no shuffling to preserve temporal order

Fields Per Record

📅 date

🏷️ asset

💰 prices

📰 news

📄 10-K / 10-Q

⚡ momentum

📈 future_price_diff (used for labeling)

Anti-Leakage Labeling

Labels derived from **future_price_diff** using train-set-computed threshold: `hold_frac × train_std`. Threshold computed **only on training data** to prevent leakage.

Methodology Overview



Feature Engineering



Sentiment Features

FinBERT — ProsusAI/finbert



3-Dimensional Output

neg

Negative

neu

Neutral

pos

Positive



Text Processing

- ✓ Split into **512-token chunks**
- ✓ **256-token stride** for overlap
- ✓ Probabilities **averaged** across chunks



Market Features

Technical Indicators



1-Day Return

Daily price change percentage



5-Day Volatility

Rolling standard deviation



Momentum Signal

Bullish=1.0, Neutral=0.0, Bearish=-1.0



14-Day Z-Scored Return

Scale-invariant across BTC and TSLA



Total Feature Dimension

Per timestep

15

features

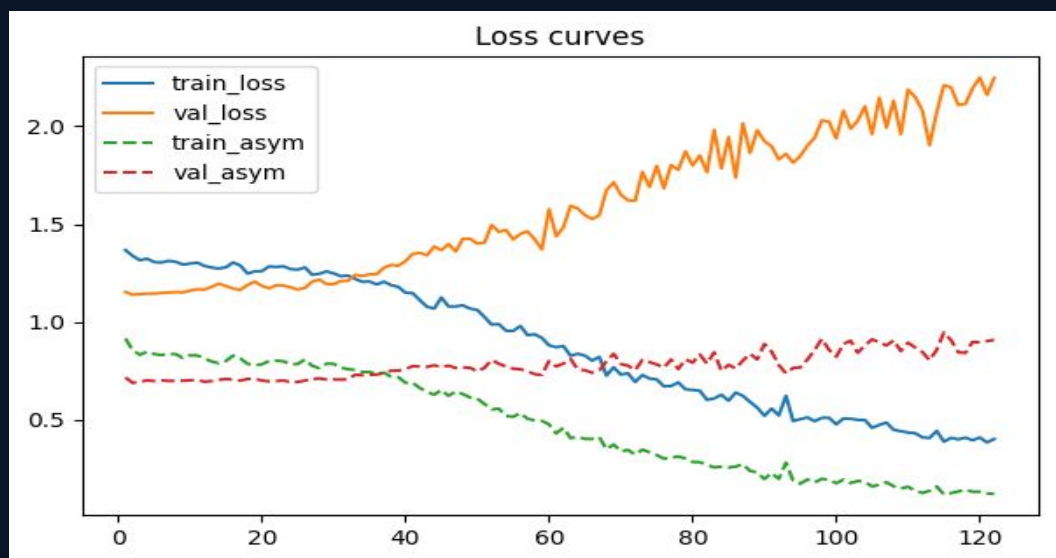
7

days window

Training & Loss Curves

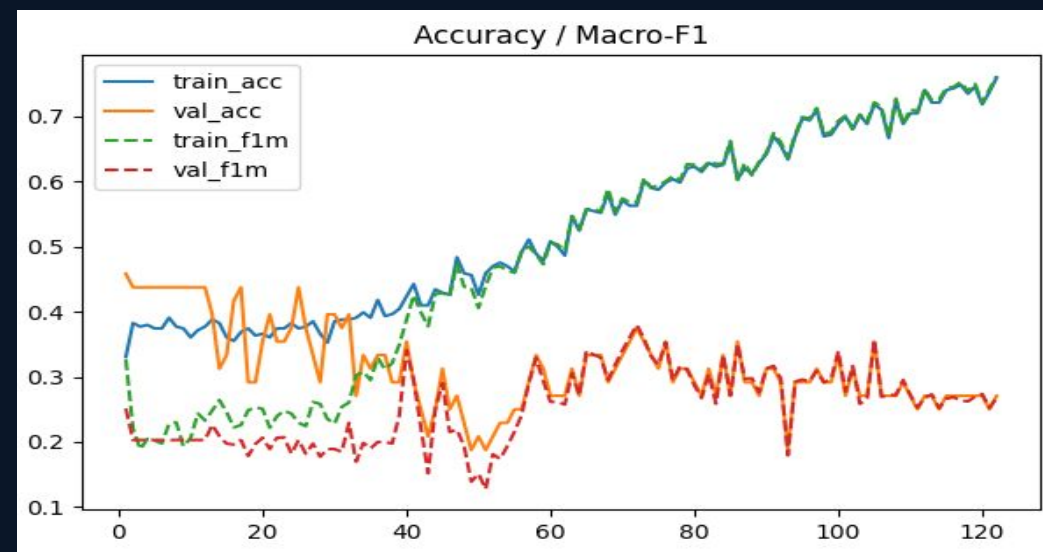
Loss Curves

Train Loss Val Loss



Performance Metrics

Accuracy Macro-F1



Batch Size

32

Max Epochs

300

Early Stopping

50

Results & Evaluation

Classification Metrics

Test Set

Accuracy 0.5

Macro F1 0.4877

Per-Class Metrics

BUY Precision / Recall

0.41

Precision

0.58

Recall

HOLD Precision / Recall

1.00

Precision

0.28

Recall

SELL Precision / Recall

0.46

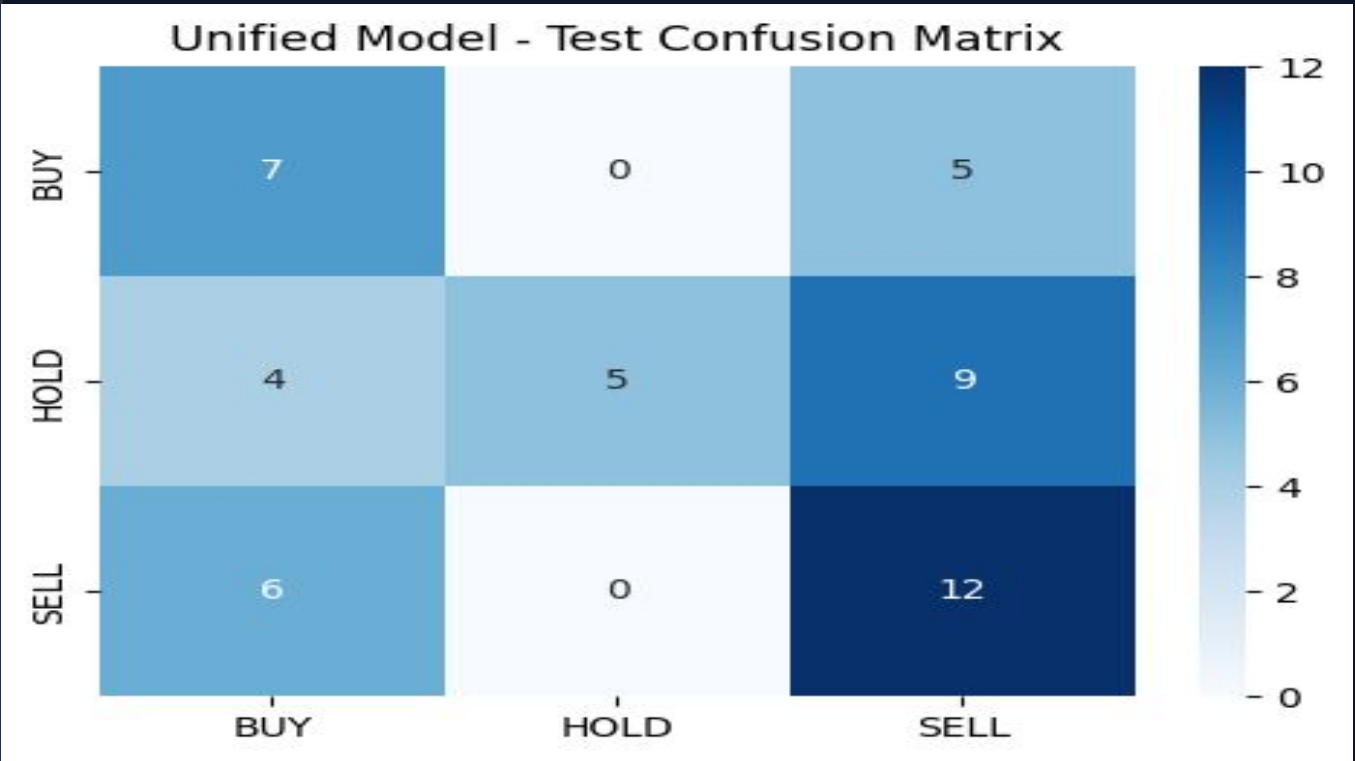
Precision

0.67

Recall

Confusion Matrix

Test Set



Post-Training Threshold Tuning

Grid search over buy_th and sell_th (0.01-0.35)

Objective: Maximize $CR + 0.2 \times SR - 0.3 \times IMDDI$

Constraint: Hold rate 15%–70% per asset

Tuned on validation set

Trading Performance & Final Predictions

Trading Metrics Test Set

Cumulative Return
19.91%

Sharpe Ratio
2.840

Max Drawdown
-7.81%

Ann. Volatility
35.88%

Compared against Buy-and-Hold baseline

Final Model
Outputs

Bitcoin (BTC)

[SELL]

Tesla (TSLA)

[BUY]

Action Signal Visualization

BUY HOLD SELL



Conclusion & Future Work



Key Contributions

1

End-to-End Pipeline

Combines NLP (FinBERT sentiment) with sequential modeling (GRU) for financial decision making

2

Asymmetric Loss Function

Aligned with real trading costs — penalizes wrong directional predictions proportional to loss magnitude

3

Unified Multi-Asset Model

Single model handles heterogeneous assets (crypto + equity) via asset embeddings



Limitations



Small Dataset

Only 237 records per asset — limited training data for deep learning models



Variable News Quality

News articles vary in relevance and reliability — noise dilutes signal



Sparse Filings

10-K and 10-Q filings often absent — missing fundamental data



Future Work



Larger Transformer Backbones

Experiment with FinGPT and other large financial language models for richer representations



Multi-Asset Portfolio Optimization

Extend to portfolio-level decision making with risk-adjusted allocation strategies